4.3 三维周期场中的近自由电子近似

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(\vec{r}) = -\frac{\hbar^2}{2m}\nabla^2 + \bar{V} + \Delta V(\vec{r})$$

$$V(\vec{r} + \vec{R}_m) = V(\vec{r})$$

- **基函数**

$$|\vec{k}> \equiv \psi_{\vec{k}}^{(0)}(\vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k}\cdot\vec{r}}$$

$$E_{\vec{k}}^{(0)} = \frac{\hbar^2 k^2}{2m} + \bar{V} \qquad \text{令 } \bar{V}=0$$

- **1阶能量修正** $E_k^{(1)} = < k|\ \Delta V(\vec{r})|\ k >= 0$

- **2阶能量修正** $E_k^{(2)} = \sum_{k' \neq k} \frac{\overrightarrow{<k'|V(r)|k>}^2}{E_k^{(0)} - E_{k'}^{(0)}}$

- 1阶波函数 $\psi_k^{(1)} = \sum_{\vec{k}' \neq \vec{k}} \frac{|\vec{k}'><\vec{k}'|\ V(\vec{r})|\ \vec{k}>}{E_{\vec{k}}^{(0)} - E_{\vec{k}'}^{(0)}}$

$$\psi(\vec{r}) = \sum_{\vec{k}} C(\vec{k}) \mid \vec{k} >= \frac{1}{\sqrt{V}} \sum_{k} C(\vec{k}) e^{i\vec{k}\cdot\vec{r}}$$

$$\hat{H}\psi(\vec{r}) = \hat{H} \sum_{\vec{k}} C(k) \mid k >= \varepsilon \sum_{\vec{k}} C(k) \mid k >$$

$$\sum_{k'} C(k') E_{\vec{k}'}^{(0)} |\ k' > + \sum_{k',n} C(k') V_n |\ k' \quad \vec{G}_n >= \varepsilon \sum_{k'} C(k') |\ k' >$$

- 左乘<k|，积分得到

$$C(k^{'})(E_{\vec{k}}^{(0)} - \varepsilon_{\vec{k}}) + \sum_{n} C(k - \overrightarrow{G_n}) V_n = 0$$

- 上述久期方程未知变量： $C(k), C(k - G_n)$的系数

- 令系数行列式为零，有定解；

- 截断至某个G_n，k附近几个倒格矢的解是比较精确的。

- **矩阵元（振幅）**

$$<\vec{k}'|V(\vec{r})|\vec{k}>=\int_{allspace} d\vec{r}\,\psi_{k'}^{(0)}V(\vec{r})\psi_{k}^{(0)}=\frac{1}{N\Omega_c}\sum_n e^{i(\vec{k}-\vec{k}')\cdot\vec{R}_n}\int_{cell} d\vec{r}e^{i(\vec{k}-\vec{k}')\cdot\vec{r}}V(\vec{r})$$

$$=\sum_n \delta_{\vec{k}',\vec{k}+\vec{G}_n}\frac{1}{\Omega_c}\int_{cell} d\vec{r}e^{i(\vec{k}-\vec{k}')\cdot\vec{r}}V(\vec{r})=\sum_n \delta_{\vec{k}',\vec{k}+\vec{G}_n}V_n$$

- **利用晶格周期性可作傅里叶展开**

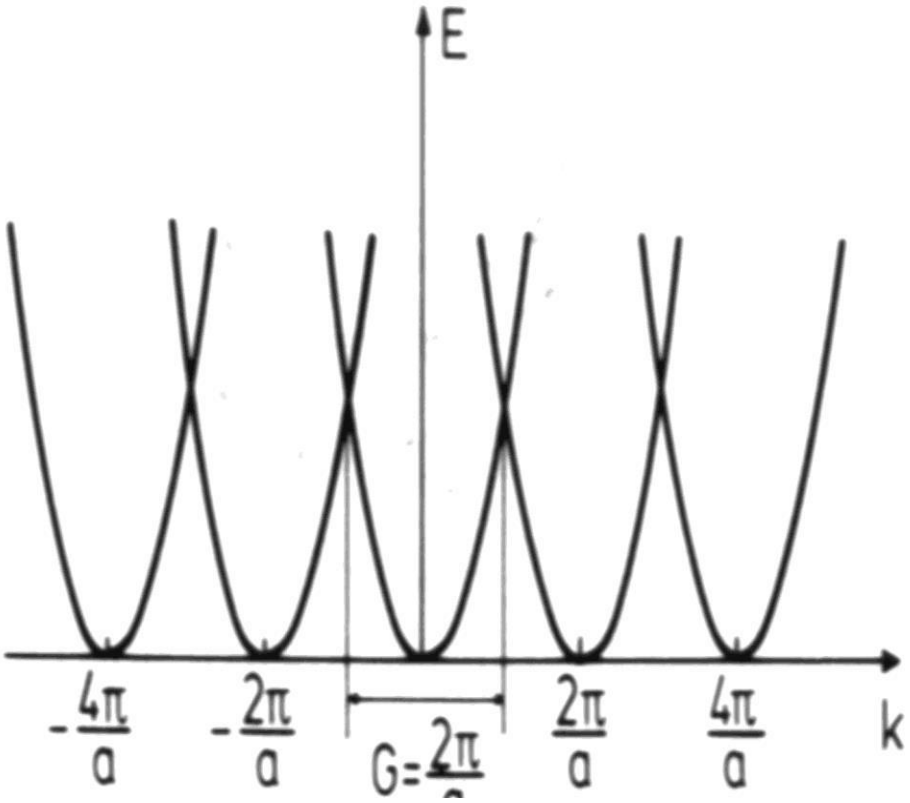
$$V(\vec{r}+\overrightarrow{R_m})=V(\vec{r})$$

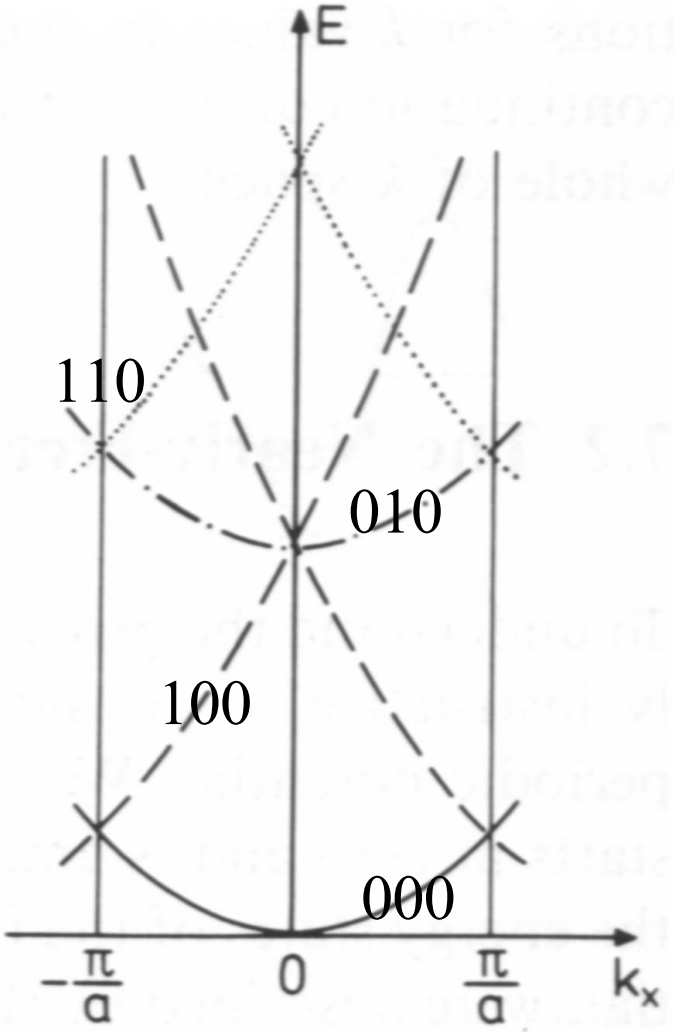
$$V(\vec{r})=\sum_n e^{i\overrightarrow{G_n}\cdot\vec{r}}V_n$$

$$V_n=\frac{1}{\Omega_c}\int_{cell} d\vec{r}e^{-i\overrightarrow{G_n}\cdot\vec{r}}V(\vec{r})=\frac{1}{N\Omega_c}\int_{allspace} d\vec{r}e^{-i\overrightarrow{G_n}\cdot\vec{r}}V(\vec{r})$$

- **周期晶格势只耦合波矢相差倒格矢的平面波，进而在布里渊区边界产生能带隙。**

一维与三维（简单立方）*k*-点示意





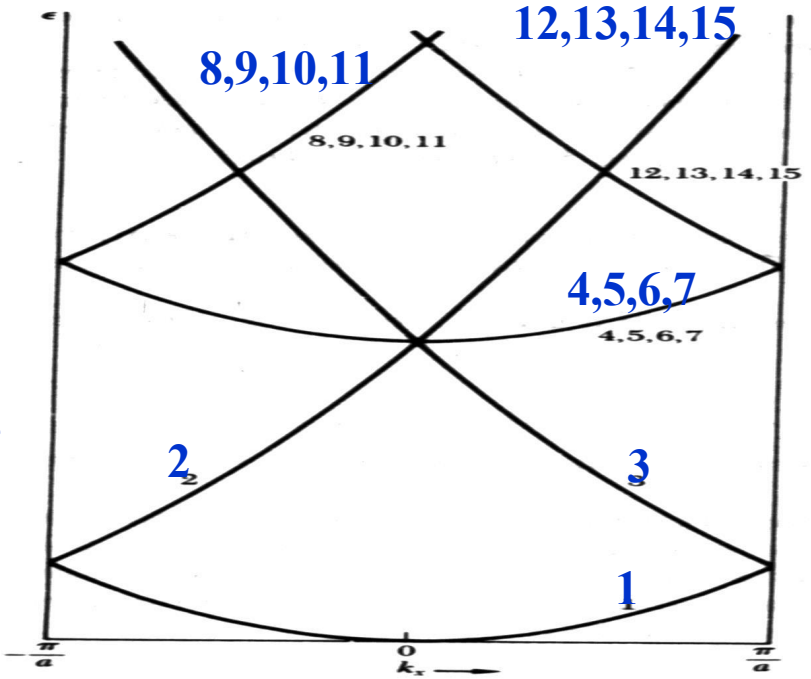
1D： *k*-点在直线分布

3D：形成规则的点阵

简单立方自由电子能带（空晶格）

$$E(k_x, k_y, k_z) = \frac{\hbar^2}{2m}[(k_x + G_x)^2 + (k_y + G_y)^2 + (k_z + G_z)^2]$$

基特尔：P126 Fig.8



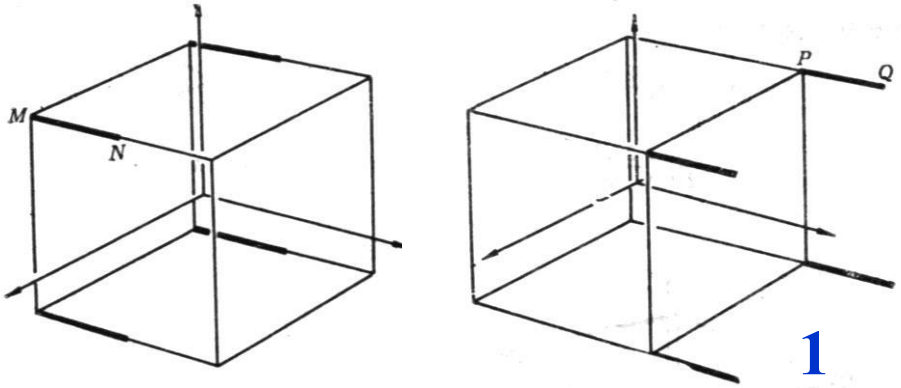
Band	$Ga/2\pi$	$\epsilon(000)$	$\epsilon(k_x00)$
1	000	0	k_x^2
2,3	$100, \bar{1}00$	$(2\pi/a)^2$	$(k_x \pm 2\pi/a)^2$
4,5,6,7	$010, 0\bar{1}0, 001, 00\bar{1}$	$(2\pi/a)^2$	$k_x^2 + (2\pi/a)^2$
8,9,10,11	$110, 101, 1\bar{1}0, 10\bar{1}$	$2(2\pi/a)^2$	$(k_x + 2\pi/a)^2 + (2\pi/a)^2$
12,13,14,15	$\bar{1}10, \bar{1}01, \bar{1}\bar{1}0, \bar{1}0\bar{1}$	$2(2\pi/a)^2$	$(k_x - 2\pi/a)^2 + (2\pi/a)^2$
16,17,18,19	$011, 0\bar{1}1, 01\bar{1}, 0\bar{1}\bar{1}$	$2(2\pi/a)^2$	$k_x^2 + 2(2\pi/a)^2$

三维面心立方晶格自由电子能带

能量单位：$\frac{\hbar^2}{2m}(\frac{2\pi}{a})^2$　　波矢单位：$\frac{2\pi}{a}$

***Band*1**　$\Gamma(000)$　$E_\Gamma = 0$　$X(010)$　$E_x = 1$

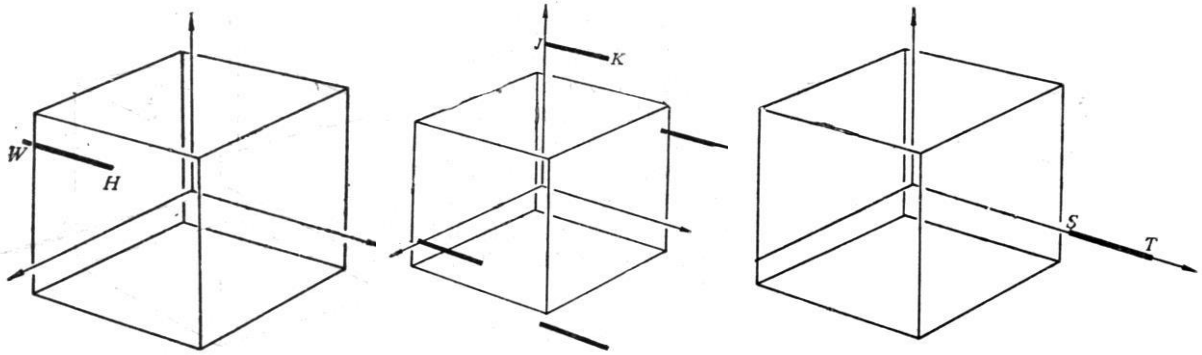
***Band* 2**　$\Gamma(1\bar{1}1)$　$E_\Gamma = 3$　$X(101)$　$E_x = 2$



1

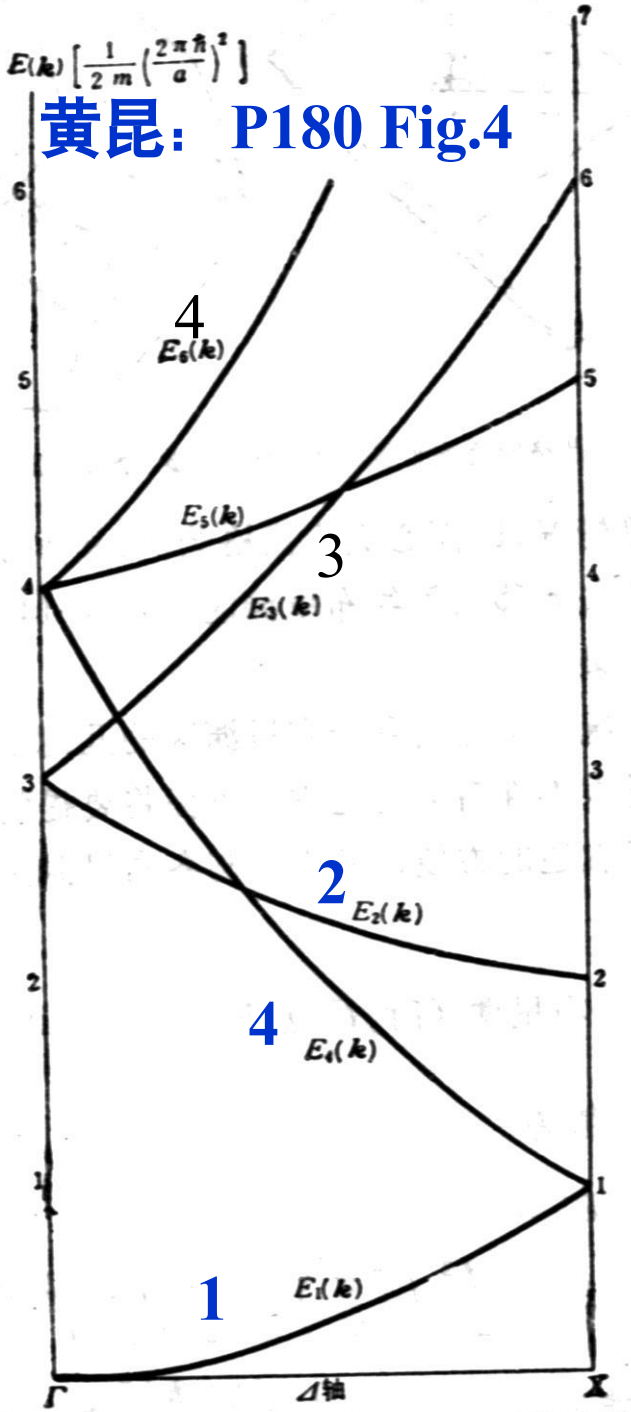
***Band* 3**　$\Gamma(111)$　$E_\Gamma = 3$　$X(121)$　$E_x = 6$

***Band* 4**　$\Gamma(0\bar{2}0)$　$E_\Gamma = 4$　$X(0\bar{1}0)$　$E_x = 1$

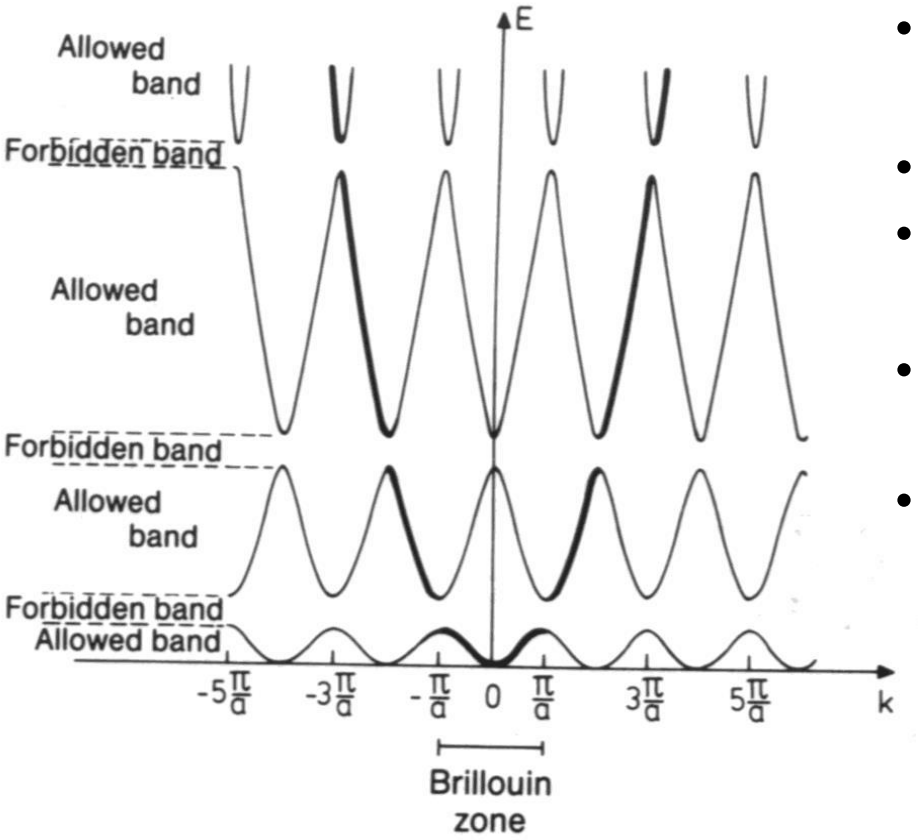


***Band* 5**　$\Gamma(002)$　$E_\Gamma = 4$　$X(012)$　$E_x = 5$

***Band* 6**　$\Gamma(020)$　$E_\Gamma = 4$　$X(030)$　$E_x = 9$



简约波矢与自由电子波矢的关系



- 简约波矢κ标记平移算符的量子态，势为任意周期势
- 自由电子波矢k只标记自由电子态，
- 常数势

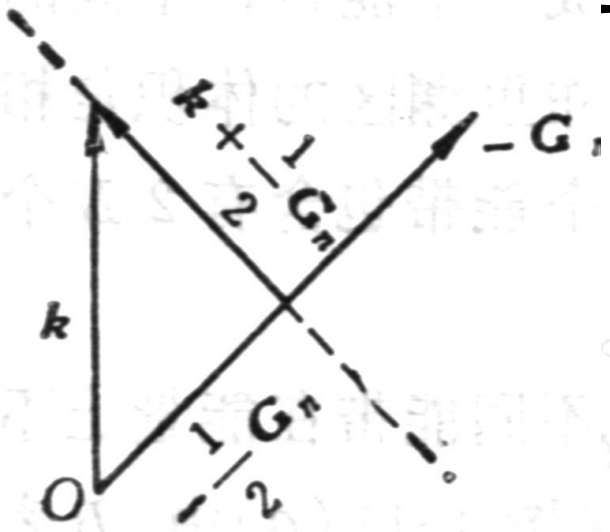
- κ限于第一布里渊区； k无此限制
- 两者关系：$k = n\dfrac{2\pi}{a} + \kappa$

➢ 不同原胞之间的相位关系

能带态的标记： n, κ

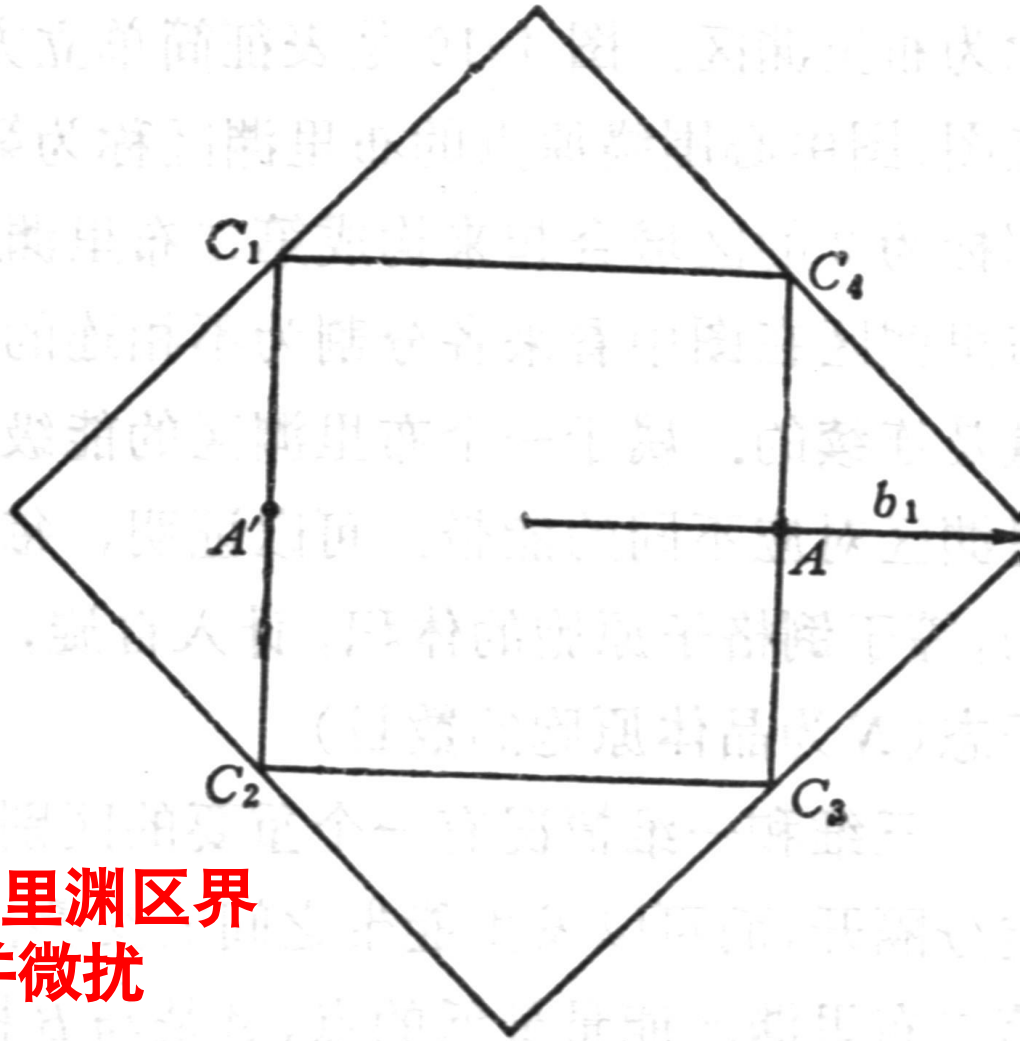
$$e^{i\kappa R} = e^{ikR}$$

一个k，可以有多个k-G，0阶能量简并

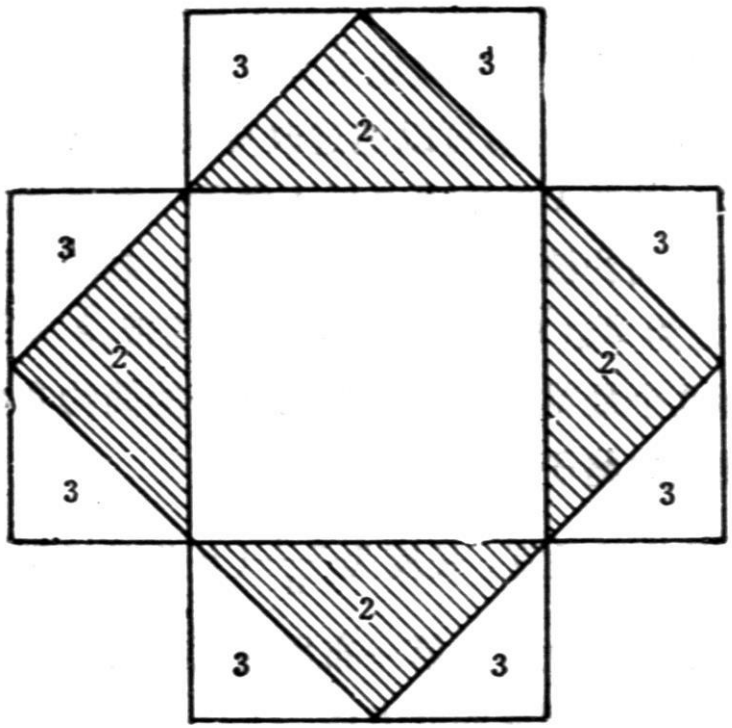


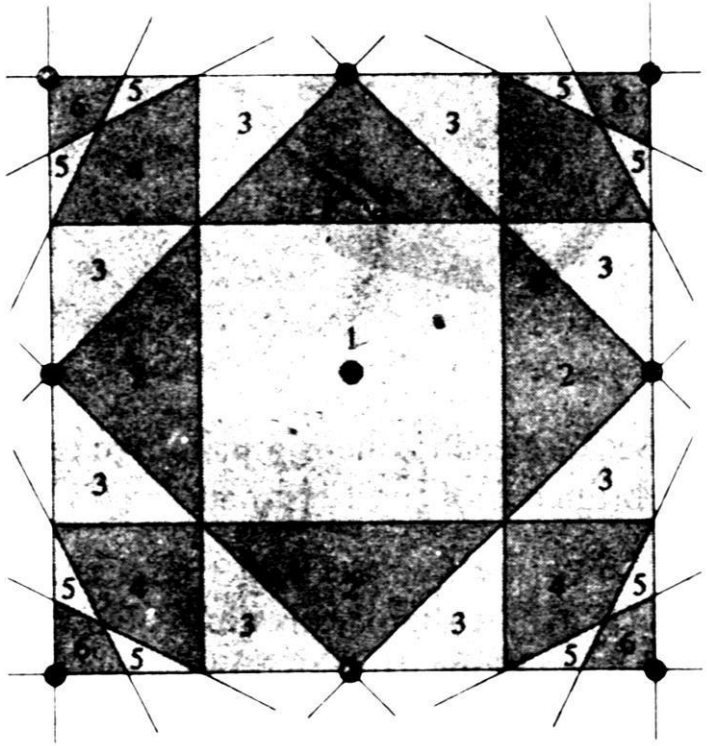
$$|\vec{k}+\overrightarrow{G_n}|^2=|k|^2$$

$$\overrightarrow{G_n}\cdot(\vec{k}+\overrightarrow{G_n}/2)=0$$



k在倒格矢的中垂面——布里渊区界面上，或在附近，要用简并微扰





不同布里渊区，$E(k)$不连续；同布里渊区，$E(k)$连续

简单立方晶格 Wigner–Seitz 元胞

$$\boldsymbol{a}_1 = \frac{a}{2}(\hat{y} + \hat{z} - \hat{x}), \quad \boldsymbol{a}_2 = \frac{a}{2}(\hat{z} + \hat{x} - \hat{y}),$$

$$\boldsymbol{a}_3 = \frac{a}{2}(\hat{x} + \hat{y} - \hat{z})$$

WS元胞为截角正八面体（如图2.4（b）），格点配位数z=8.

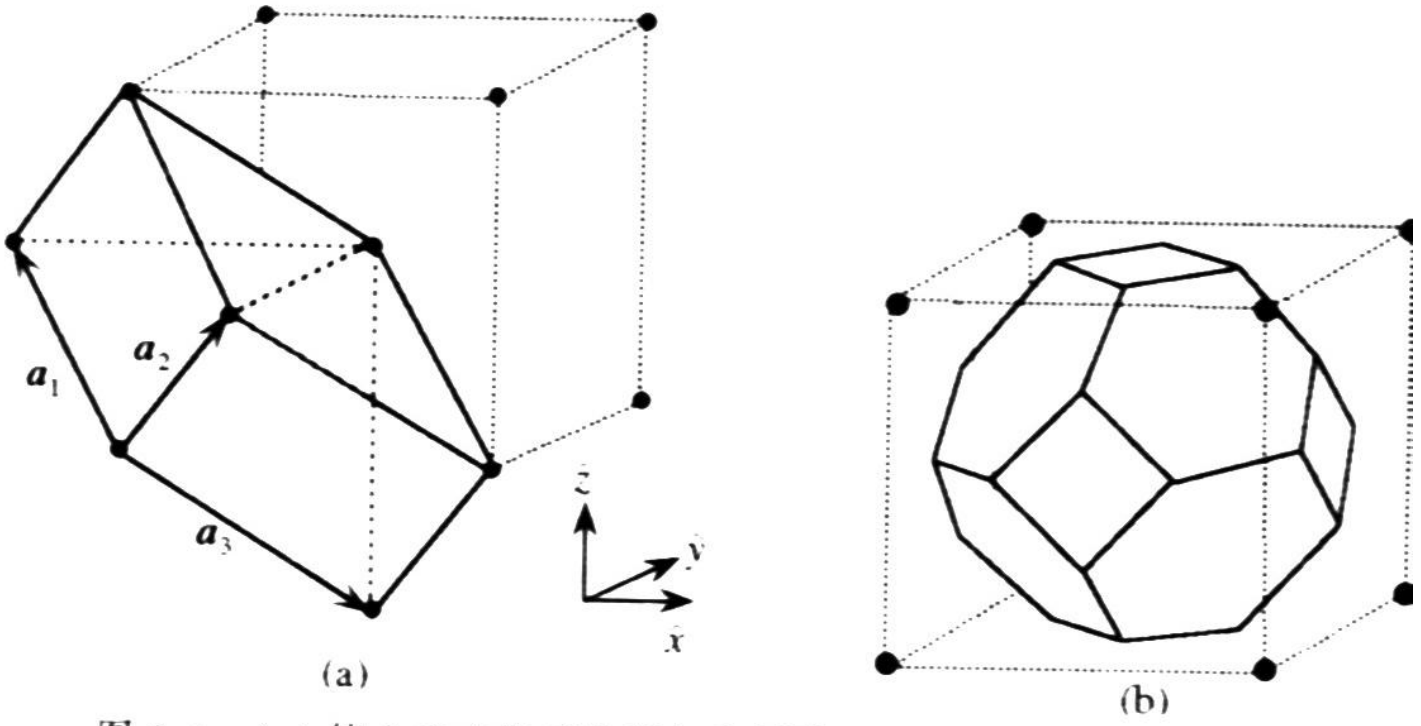


图 2.4 （a）体心立方格子的基矢和原胞（b）体心立方格子的 WS 原胞

FCC 晶格 Wigner–Seitz 元胞

$$\boldsymbol{a}_1 = \frac{a}{2}(\hat{y}+\hat{z}), \quad \boldsymbol{a}_2 = \frac{a}{2}(\hat{z}+\hat{x}), \quad \boldsymbol{a}_3 = \frac{a}{2}(\hat{x}+\hat{y})$$

WS元胞为正十二面体（如图2.5），格点配位数z=12.

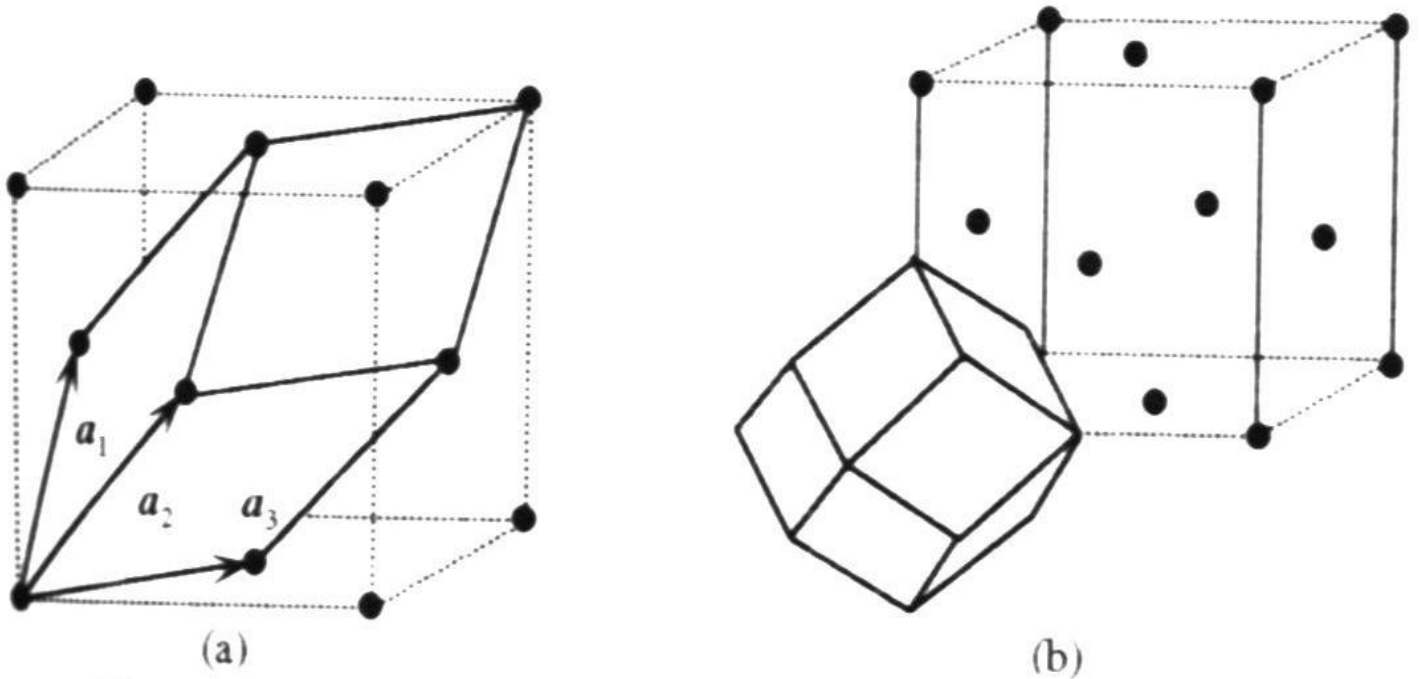
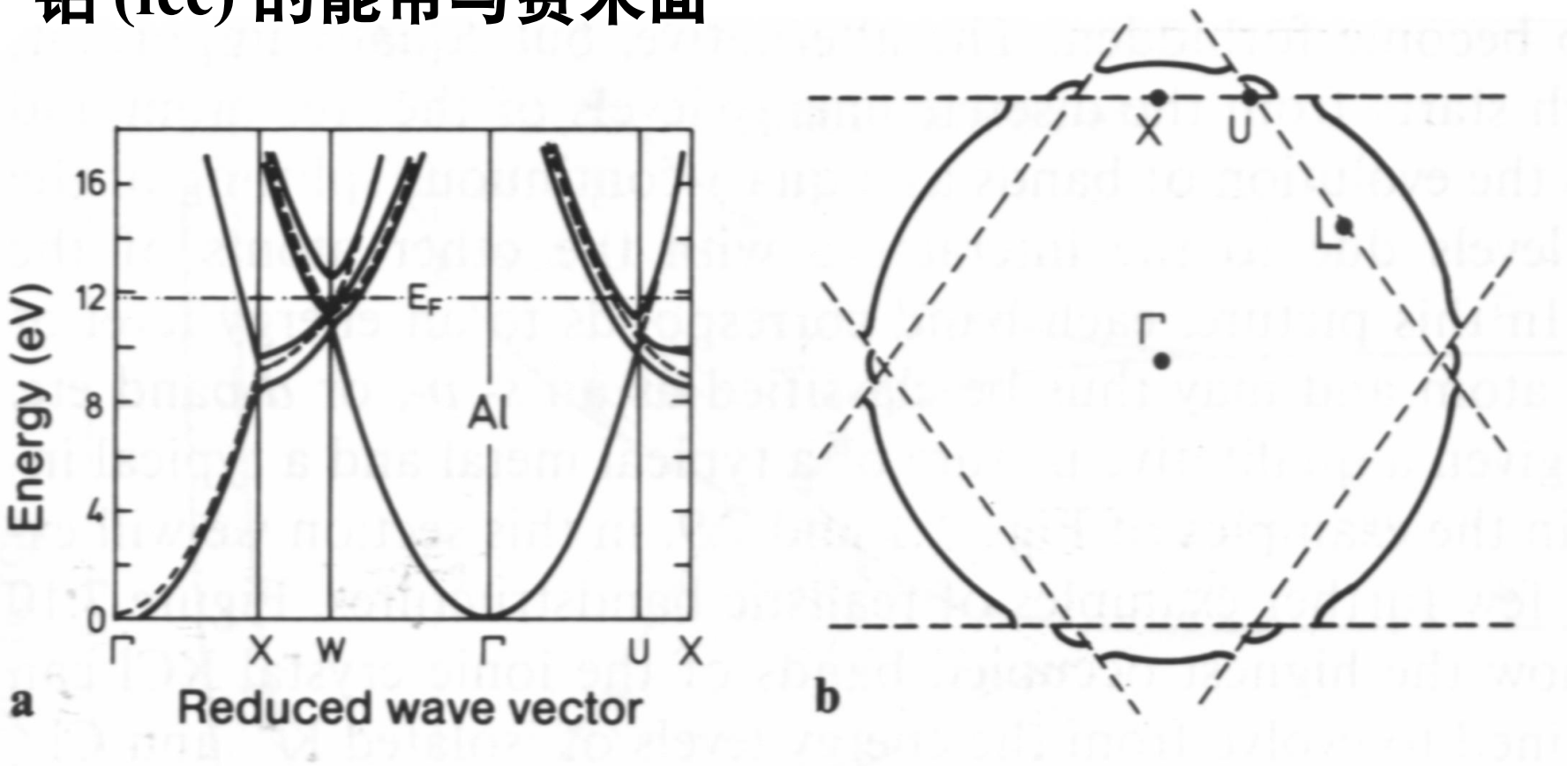


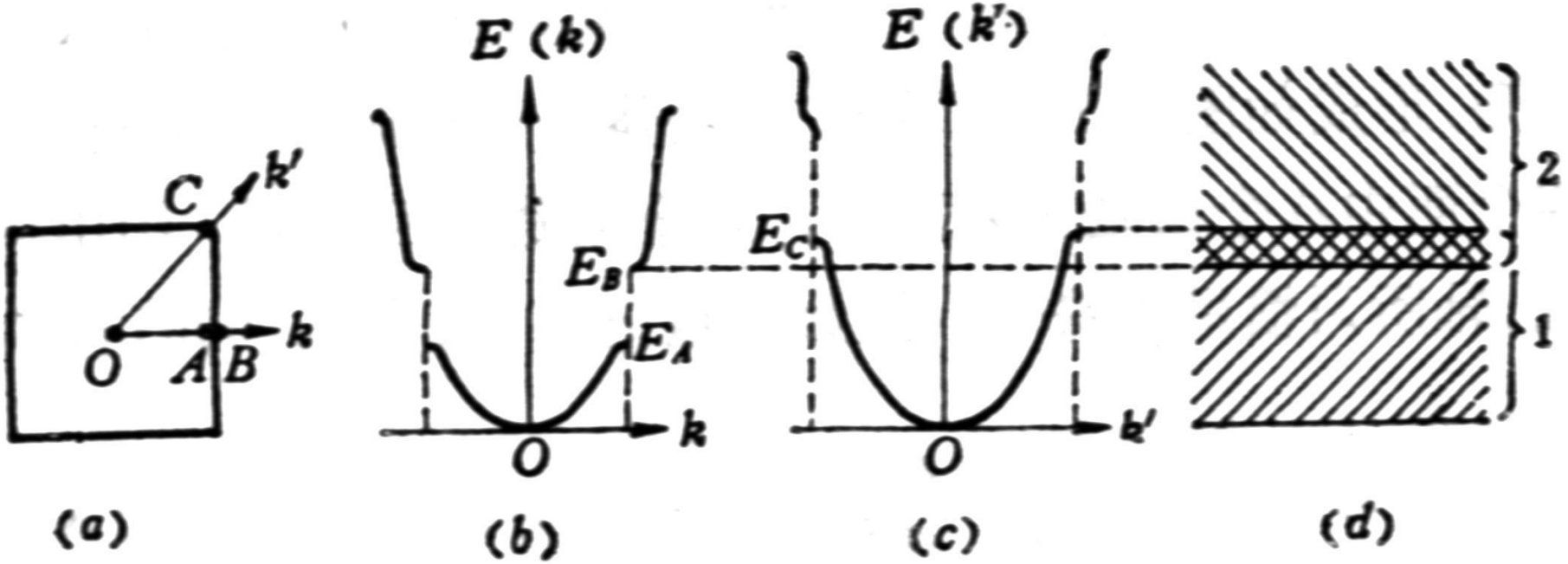
图 2.5 （a）面心立方格子的基矢和原胞（b）面心立方格子的 WS 原胞

铝 (fcc) 的能带与费米面



a. Al**的能带（虚线是自由电子结果）**; b. Al（fcc)**的**Brillouin**区的截面。虚线是第一**Brillouin**区边界，实线是费米面。**

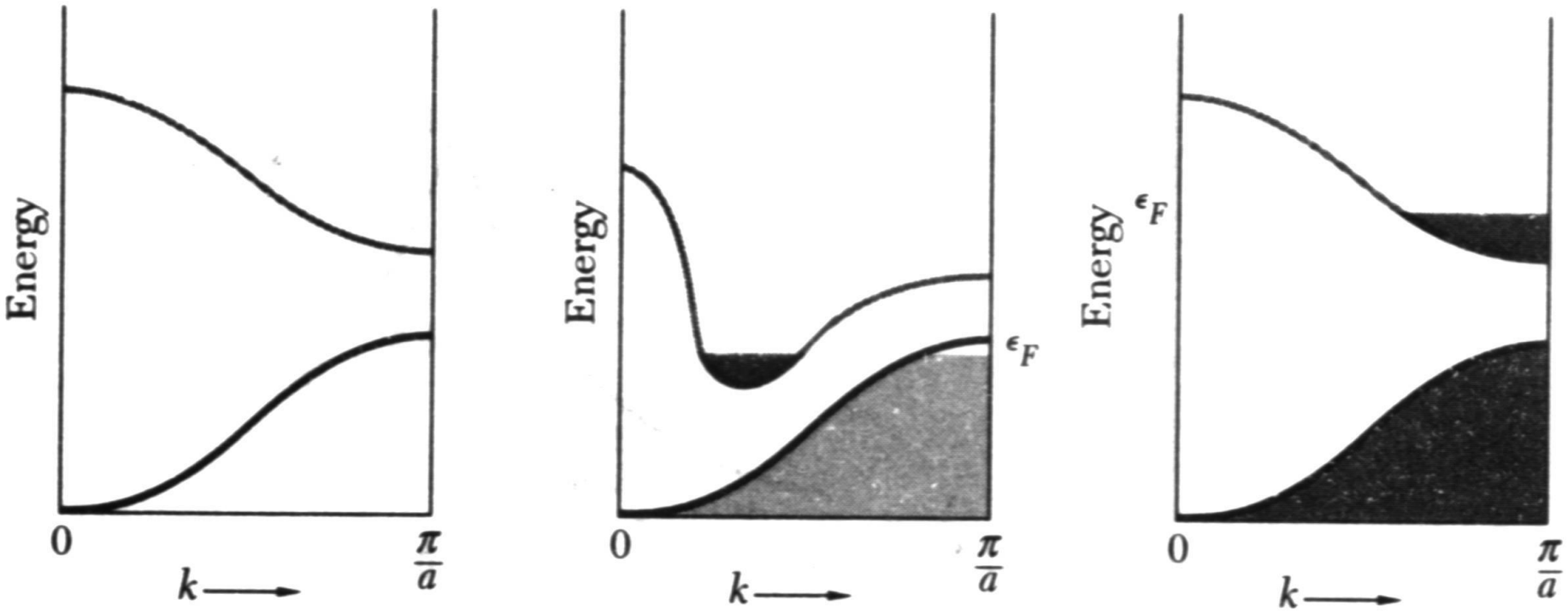
三维能带交叠



三维不同于一维，不同能带可以交叠

能带交叠图示

基特尔：P128 Fig.11



被占据态的能带结构示意图：(a)绝缘体（b）由于能带交叠得到的金属或半金属（c）由于电子浓度而形成的金属。在(b)中的交交叠不一定发生在布里渊区内的一个方向上。如果交叠小，只涉及比较小的态，便称为半金属。